

Week 2: Deciding under *fire*

Context Triage, uncertainty, and decisions under scarcity
Engineering Building for *Broken* Environments

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When there is not enough time, information, or capacity to treat everyone, who should we help FIRST?

- Who is most at risk, and how do we decide *quickly*?
- How can we ensure that *quick* decisions are based on careful thought?
- How can a hospital expand, coordinate, and remain accountable to millions of *quick* decisions under surge?

Deciding Under Fire: Triage, Uncertainty & Care

Lecturer: *Prof. Annie Hartley*

Light

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Outline

- 01 The problem: need > > > capacity
- 02 The birth of Triage: *Larrey's "flying ambulances"*
- 03 Triage: Assessment, reassessment, handover
- 04 Pre-positioning
- 05 Deciding under fire: Checklists vs AI **ENGINEERING**

The solution

Triage: Sorting and prioritizing patients according to assessed urgency

patient

noun

A person receiving medical care.

From Latin *patiens*: suffering, enduring.

patiente

French · imperative of patienter

“Wait. Be patient.”

Suffering $f(\text{Time})$

What is a priority?

$$\text{Priority} = \frac{\text{Urgency} \times \text{Benefit gained}}{\text{Time required} \times \text{Resources required}}$$

Speed at which a delay causes harm

Terminology

Opportunity cost

The benefit forgone when scarce resources go to one patient rather than another.

What does this decision make unavailable to someone else?

Triage...black and white?



One operating theatre. Who goes first?

by Mary-Anne Hartley · just now

Make a choice:

- ☐ An 8-month-old baby: 90% chance of survival with surgery now; 60% if delayed.
- ☐ An 80-year-old woman: 70% chance of survival with surgery now; 40% if delayed.

Vote →

Show results

Share



LiGHT

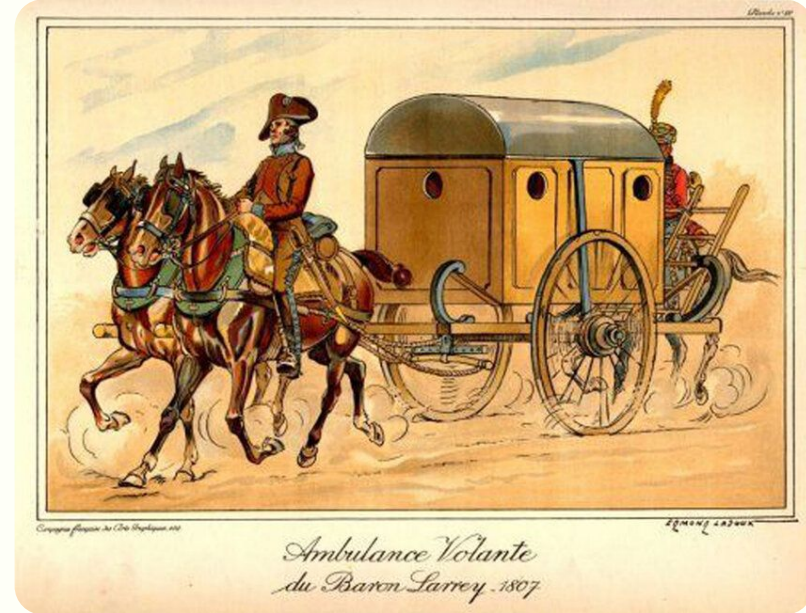
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Larrey's flying ambulances, 1766-1842

Mobile medical teams in fast horse-drawn vehicles to respond to the wounded in war.

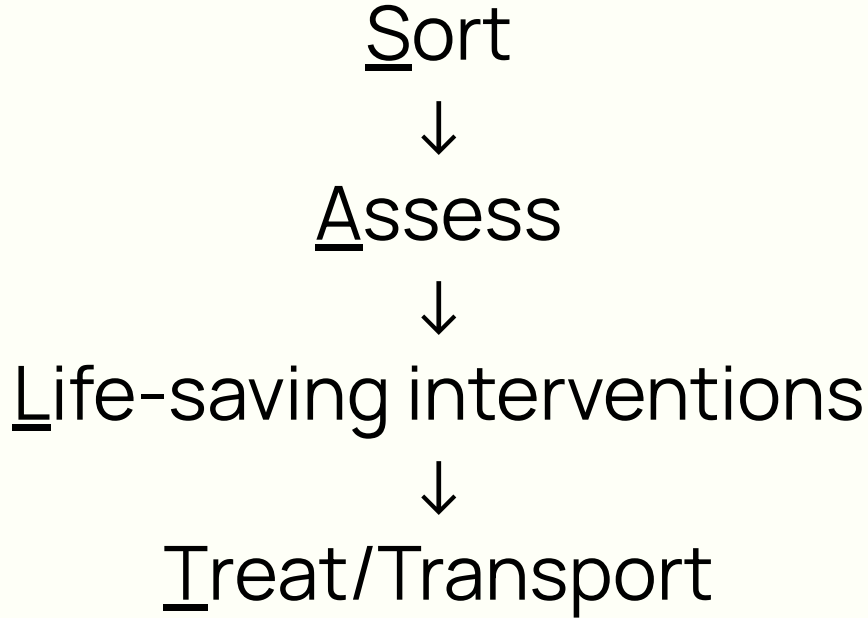
But what were they optimizing?

- Treat the most severely injured first?
- Treat those most likely to survive?
- Treat those who can return to duty fastest?
- Treat people in arrival order?
- Or preserve resources for patients who appear more salvageable?



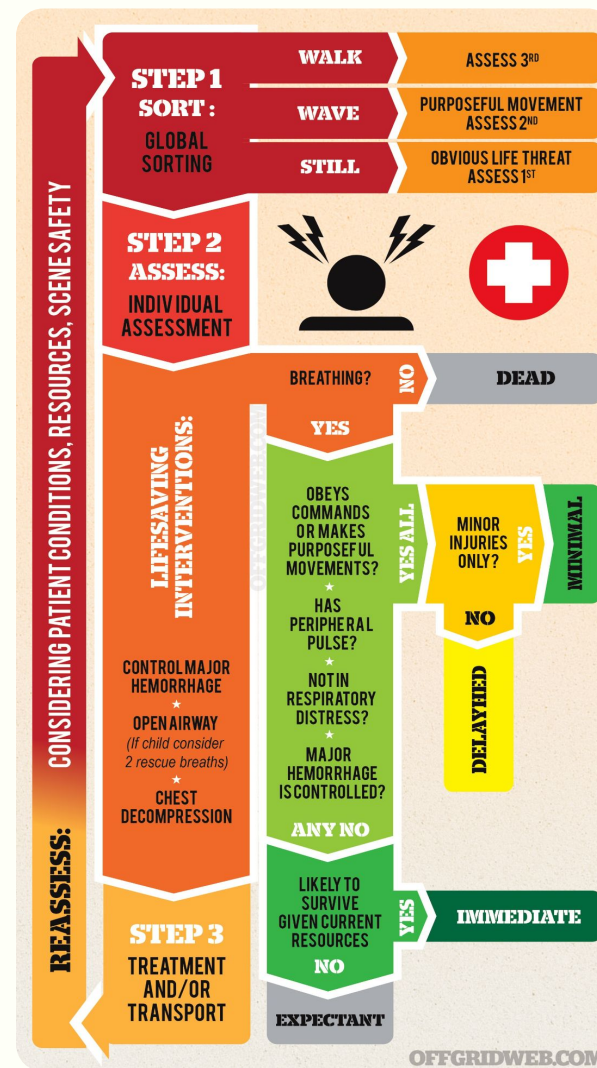
Standardized triage system

SALT



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CONTEXT

Standardized triage system

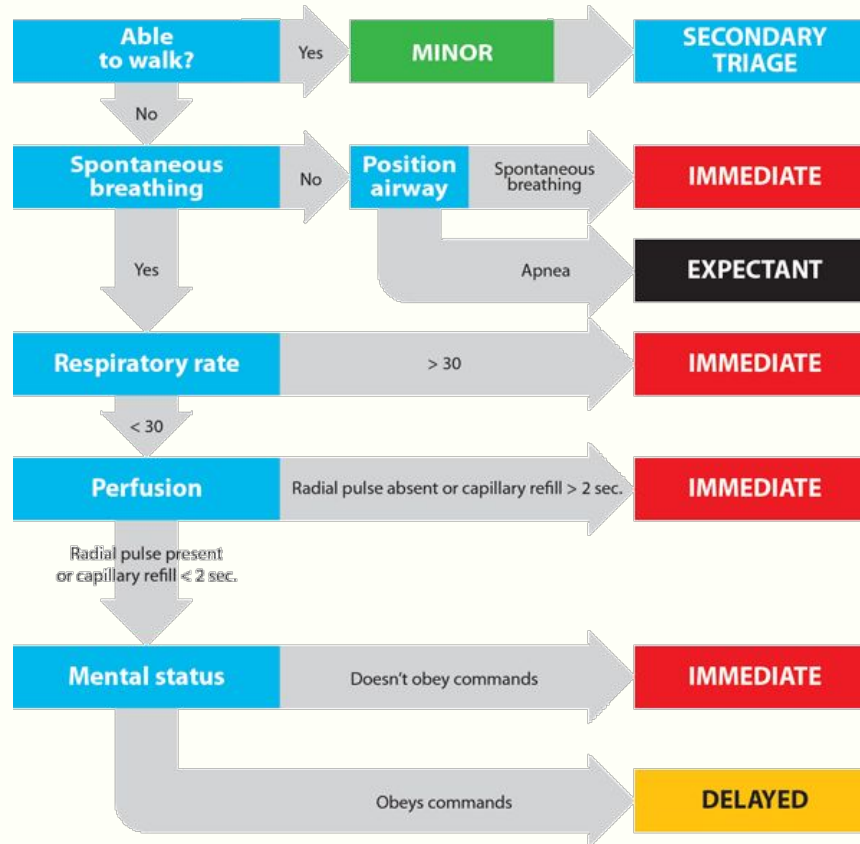
START

Simple Triage And Rapid Treatment

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Figure 1: START method for adults



CONTEXT

Terminology

Expectant

Categorization for a patient who is unlikely to survive given the available time and resources

START



Images made with prompting to chatGPT

Terminology

P-category

- **P0:** Deceased
- **P1:** URGENT and manageable given available resources
- **P2:** Can wait, next up
- **P3:** Walking wounded
- **P4:** Expectant “left to die”



START...context agnostic



Images made with prompting to chatGPT

Terminology

P-category

- **P0:** Deceased
- **P1:** URGENT and manageable given available resources
- **P2:** Can wait, next up
- **P3:** Walking wounded
- **P4:** Expectant “left to die”

Assessment and reassessment parameters

ABCDE... F... G

A structured assessment—and a reminder

A	Airway	Can air get in?
B	Breathing	Is oxygen moving into the body?
C	Circulation	Is blood moving around the body? Is there major bleeding?
D	Disability	What is happening neurologically—consciousness, pupils, seizures, blood glucose?
E	Exposure / environment	Look for injuries, rashes, burns, bleeding and temperature problems, while preserving dignity and preventing hypothermia.
F	Did I forget something...?	Ah yes...
G	GLUCOSE!	

Handover

SOAP

S — Subjective
O — Objective
A — Assessment
P — Plan

S	Situation What is happening right now?	"I'm calling about a 26-year-old with abdominal trauma. His blood pressure is falling despite fluids."
B	Background What is important to the context?	"He was stable on arrival."
A	Assessment What do I think is going on?	"He is now pale and confused. I'm concerned about internal bleeding."
R	Recommendation / Request What do I need from you, and how urgently?	"I need you in resuscitation now and would like the theatre alerted."

NOTICE: more mnemonics...more checklists for BASIC things that we clearly already know
"decision making tools"

Triage...and then...?

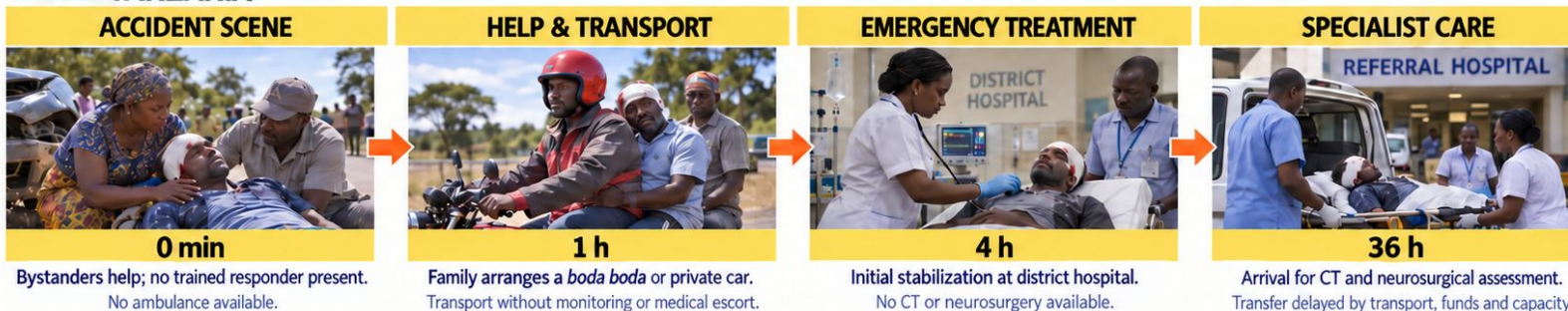
Referral chain

Triage decisions must also be based on upstream “pre-positioned” resources

URBAN SWITZERLAND



RURAL TANZANIA



Rapid decisions do not rely on rapid statistics

Florence Nightingale

British nurse based in Constantinople during the Crimean war (1853-1856)



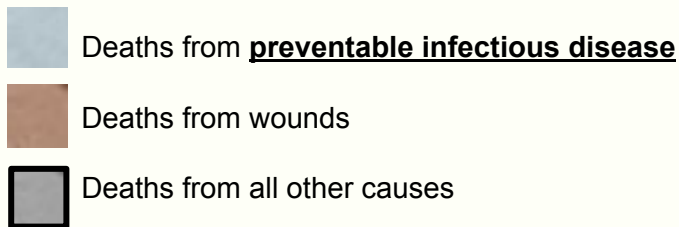
Rapid decisions do not rely on rapid statistics

Florence Nightingale Advocacy through evidence

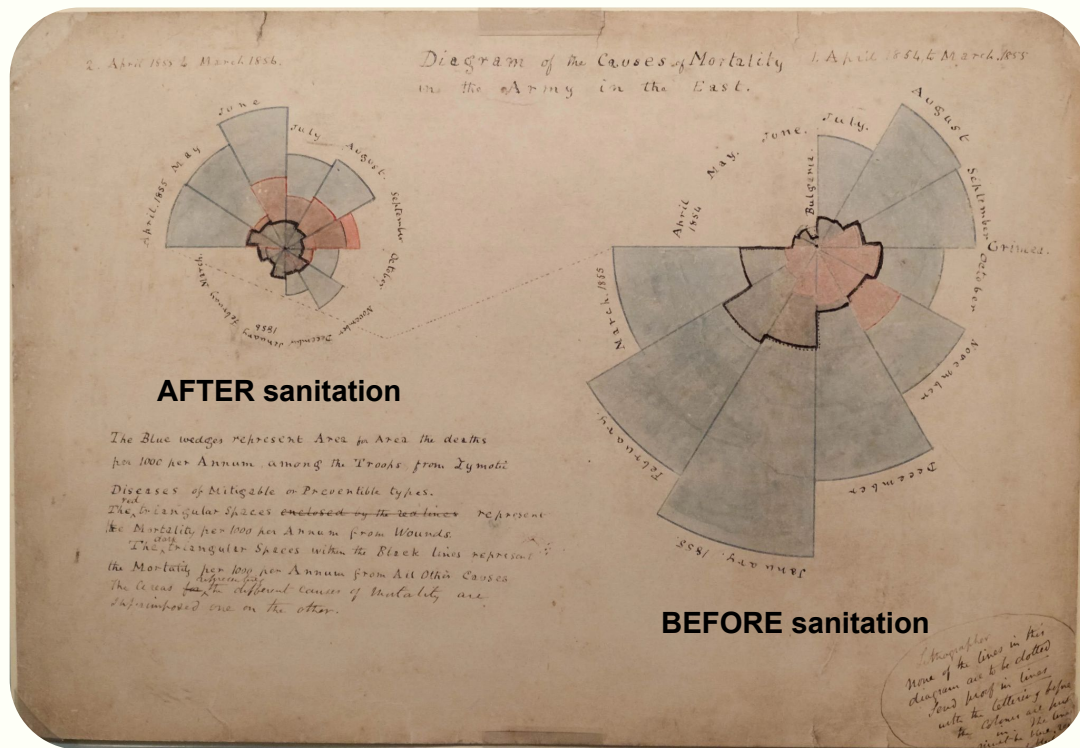
- Self-taught statistician
- Conducted one of the first **interventional before-and-after** studies on the value of sanitation
- Triage based on assessment...and reassessment according to resources



Pre-positioning of statistical reasoning



1 wedge = 1 month



Pre-positioning resources

Video tour: [UNICEF Supply Division Warehouse Tour](#)



UNICEF emergency warehouse in Copenhagen

Pre-positioning resources

Video:

[UNICEF menstrual hygiene products in emergencies](#)



Inside a menstrual hygiene kit

Ok...but how can AI help?

CONTEXT

ENGINEERING

Rapid decisions are intricately complex

Problem: Complex decisions under limited reasoning budget

Solutions:

- Deterministic scaffolding (e.g. checklists)
- Externalization of memory
- Shared state (e.g. training and standardization)
- Adaptive reasoning (e.g. delegate deeper thought on more difficult tasks)

Terminology

Deterministic

Fixed rules:

same input
=
same output

Ok...but how can AI help?

Example of deterministic scaffolding

CONTEXT

ENGINEERING

SIGN IN

Anaesthetist and assistant

(Before induction of anaesthesia)

Patient confirmation

- Name
- Procedure
- Site marked

Allergies?

Anaesthetic safety

- Airway plan
- Antibiotics

E-issue blood available?



Stop before
you block

TIME OUT

Introductions

Team: Confirm patient's

- Name
- Procedure
- Site and side
- Imaging
- Allergies

Surgeon

- Significant blood loss?

Team

- Diathermy on
- Antibiotics given
- Specimen plan
- Warming on
- VTE prophylaxis on

Anyone not happy to start?

SIGN OUT

Whole theatre team

- Procedure recorded as...
- Counts correct
- Specimens labelled
- Packs removed / labelled
- Lines flushed
- Equipment problems

Post op plan

- Specific concerns?
- VTE and antibiotic plan
- Daycase?

Terminology

WHO

Surgical
Safety
Checklist

SIGN IN
TIME OUT
SIGN OUT

Ok...but how can AI help?

CONTEXT

ENGINEERING

C'mon...do we really need checklists?

Elderly man has wrong leg amputated at Austrian clinic

Freistadt Clinic apologises for 'tragic mistake' which local media said the patient did not initially recognise because of his illness

U.S. NEWS

Florida doctor faces manslaughter charge for allegedly removing wrong organ during surgery

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Ok...but how can AI help?

CONTEXT

ENGINEERING

A good checklist

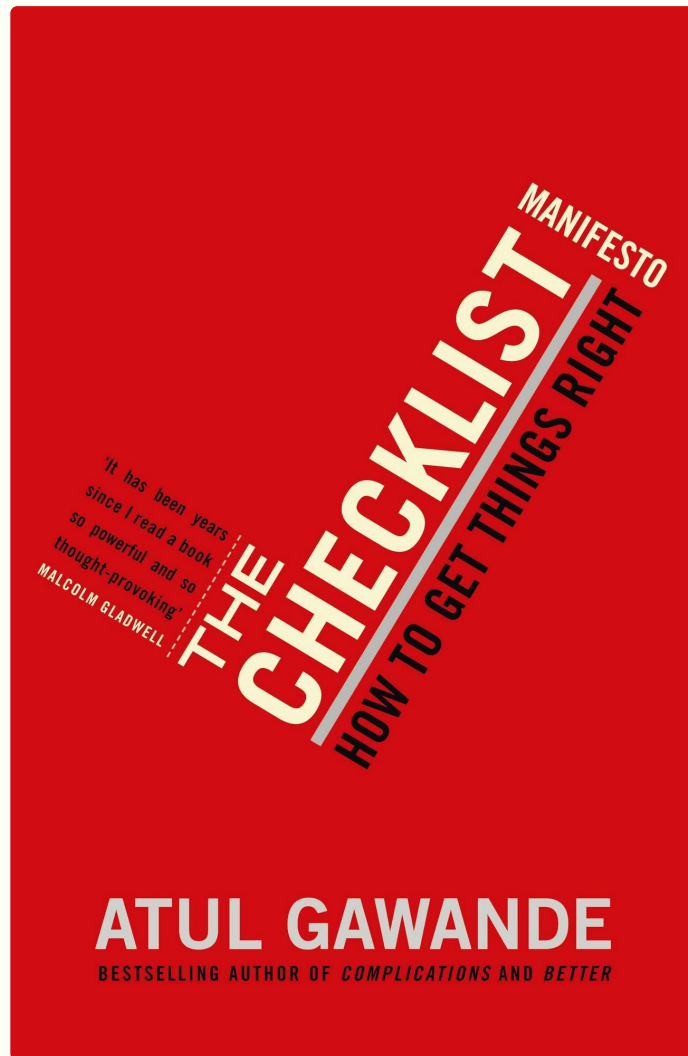
A good checklist:

1. **Protects against simple-but-preventable omissions**
2. **Creates a pause point**
3. **Turns individuals into a team**

How would you use AI here?

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End

DOMAIN

Questions?

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Building for *Broken* Environments

Lecturer: *Dr. Lars Klein*

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Engineering

This course will be heavily project-driven

Imagine that you're building for people in need, that your engineering affects them immediately

One key term here is **Accountability**

How do we prevent mistakes?

Why does this matter so much?

In the Wild...

Let's assume you're a bio-informatician...

Human genes

Mouse RIKEN IDs

MARCH1
SEPT2
DEC3

2310009E13
0610009E02

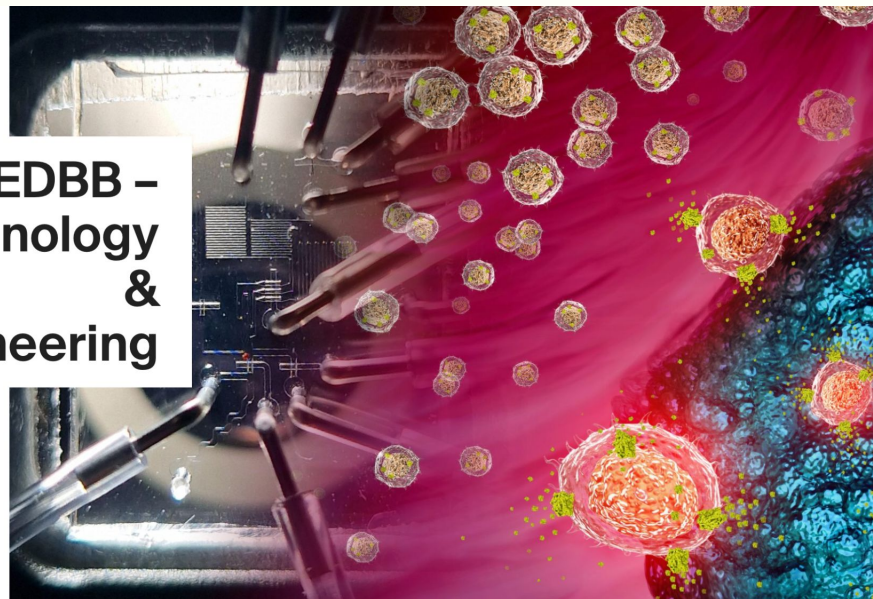
AND
OR
NOT

Drosophila genes

**EDBB –
Biotechnology
&
Bioengineering**

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In the Wild...

Mistaken Identifiers: Gene name errors can be introduced inadvertently when using Excel in bioinformatics [paper](#)

Background *When processing microarray data sets, we recently noticed that some gene names were being changed inadvertently to non-gene names.*

Results *A little detective work traced the problem to default date format conversions and floating-point format conversions in the very useful Excel program package.*

*The **date conversions** affect at least **30 gene names**;*

*the **floating-point conversions** affect at least **2,000** if Riken identifiers are included.*

*These conversions are **irreversible**; the original gene names cannot be recovered.*

In the Wild...

Gene name errors are widespread in the scientific literature [paper](#)

*Of the selected journals, the proportion of published articles with Excel files containing gene lists that are **affected by gene name errors is 19.6 %**. [...]*

Journals that had the highest proportion of papers with affected supplementary files were Nucleic Acids Research, Genome Biology, Nature Genetics, Genome Research, Genes and Development and Nature (>20 %)

In the Wild...

Guidelines for human gene nomenclature [paper](#)

Box 3 Scenarios that may merit a symbol change

...

- **Symbols that affect data handling and retrieval.** For example, all symbols that autoconverted to dates in Microsoft Excel have been changed (for example, *SEPT1* is now *SEPTIN1*; *MARCH1* is now *MARCHF1*); tRNA synthetase symbols that were also common words have been changed (for example, *WARS* is now *WARS1*; *CARS* is now *CARS1*).

[Show less](#) ^

In the Wild...

Gene name errors: Lessons not learned [paper](#)

Although recent changes have been made to human and mouse gene names to prevent conversion to dates [3], it is uncertain whether such changes have propagated through to other species. To assess this, we downloaded all eukaryotic gene names available in Ensembl and imported this into Excel and collected all genes that were converted to dates

The proportion of publications with Excel gene lists that contain errors was 30.9%; substantially higher than previously reported

So now what?

[Commentary blog post from the authors](#)

Our research shows autocorrect errors, particularly in Excel spreadsheets, can also make a mess of gene names in genetic research. We surveyed more than 10,000 papers with Excel gene lists published between 2014 and 2020 and found [more than 30%](#) contained at least one gene name mangled by autocorrect.

This research follows our 2016 study that found [around 20%](#) of papers contained these errors, so the problem may be getting worse.

We believe the lesson for researchers is clear: it's past time to stop using Excel and learn to use more powerful software.

It's past time to stop using Excel and learn to use more powerful software...

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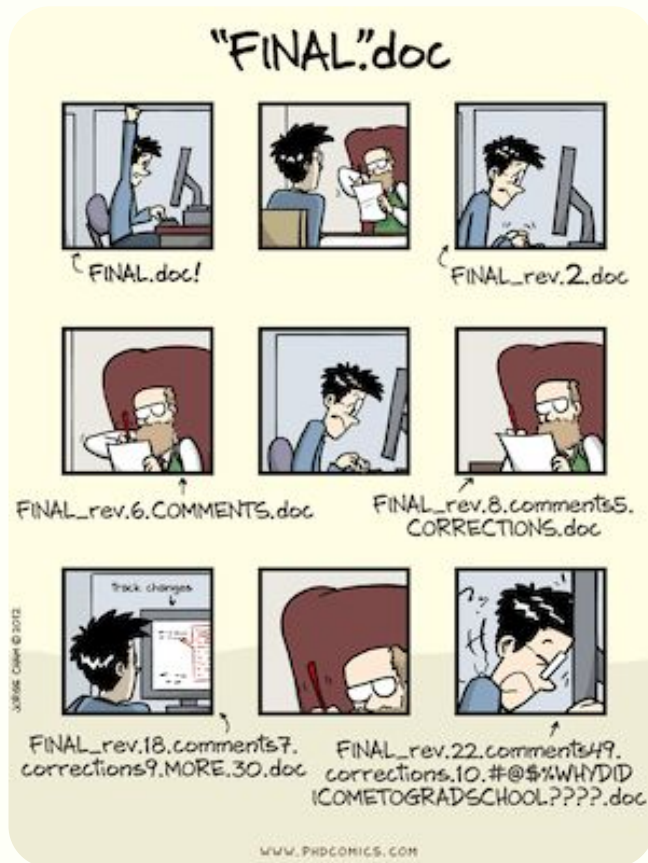
Chances are you already do



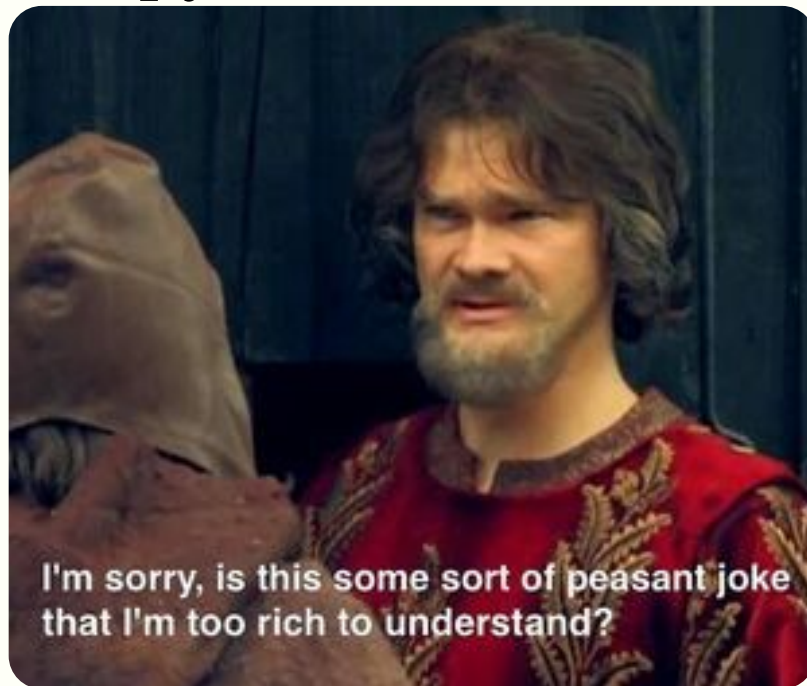
Outline

- 01 Git
- 02 Code Hygiene
- 03 Reproducibility
- 04 Tracking
- 05 Project Time!

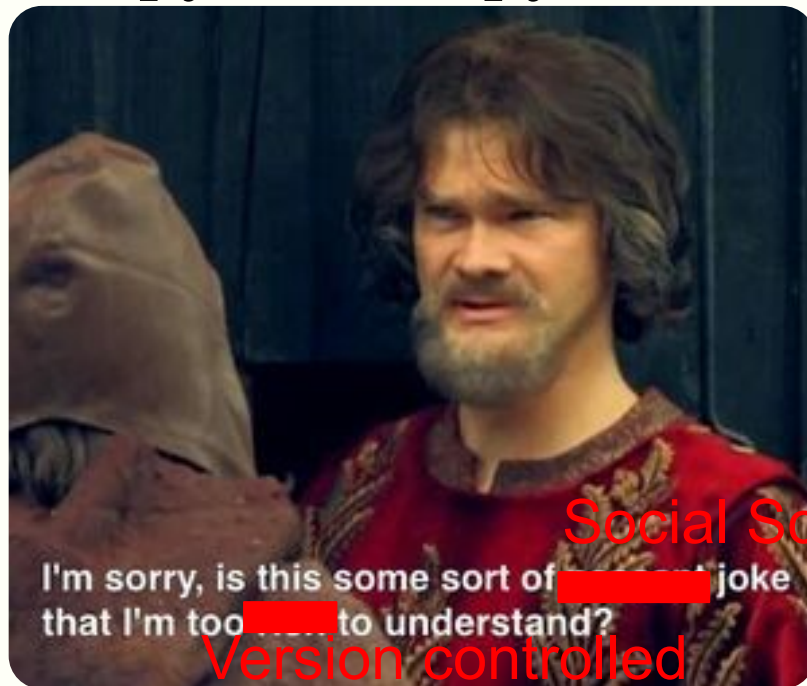
final_v2_ACTUAL.py



final_v2_ACTUAL.py



~~final_v2_ACTUAL.py~~ -> final.py



Social Sciences

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 final.py

applying feedback from ...

1 minute ago



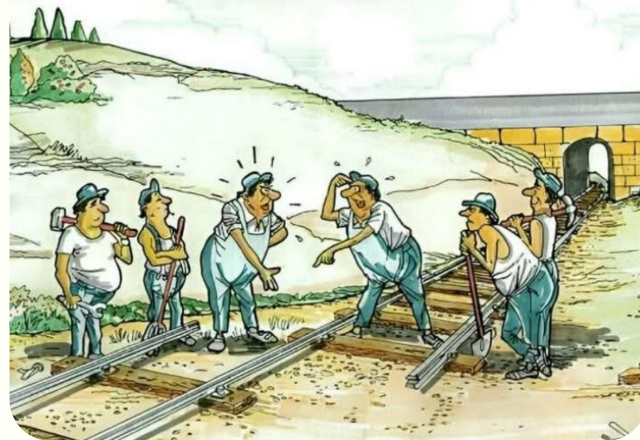
Social Sciences

Git

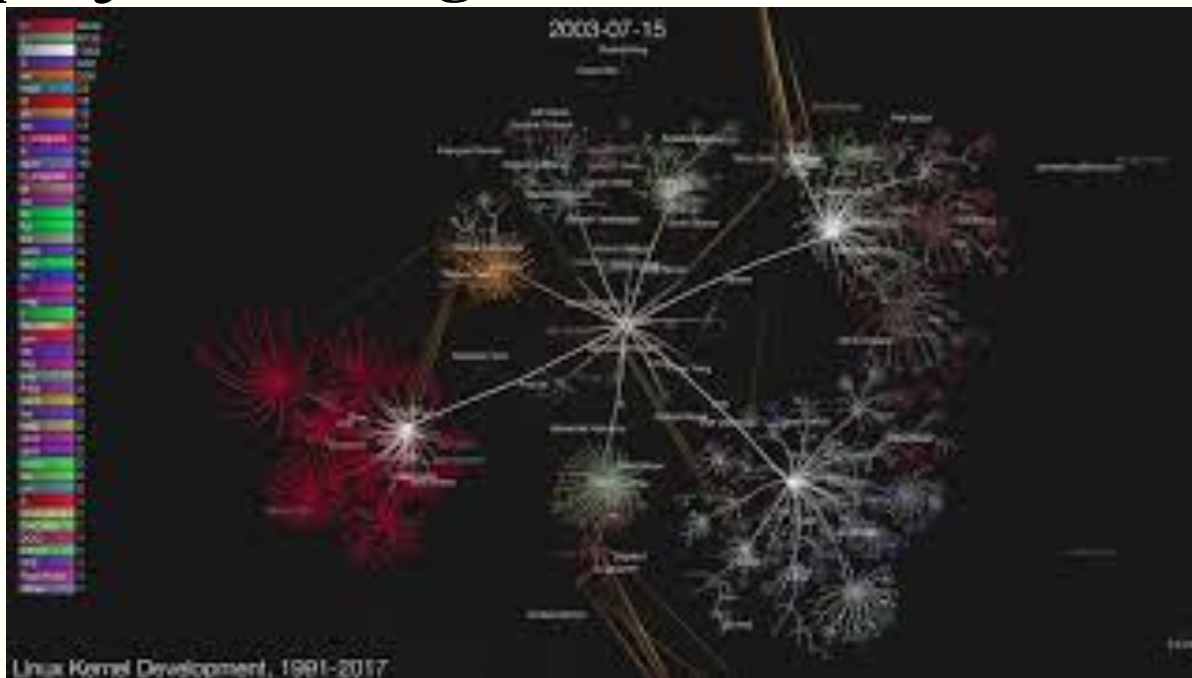
Please ask yourself:

- What is the LARGEST team size you've ever worked in?
- Did that work well?
- How much did you need to coordinate with your teammates?

**"TEAMWORK WITHOUT
COORDINATION LEADS
TO CONFUSION"**



LARGE projects with git ...



Git sanity checklist

All your projects MUST live on git → [BiGHT - Course projects GitHub](#)
This checklist is grade relevant.

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1. Never edit the main branch directly. Development happens on feature branches

Managing a branch protection rule

You can create a branch protection rule to enforce certain workflows for one or more branches, such as requiring an approving review or passing status checks for all pull requests merged into the protected branch.

Who can use this feature?

- 🔑 People with admin permissions or a custom role with the "edit repository rules" permission to a repository can manage branch protection rules.
- 📁 Protected branches are available in public repositories with GitHub Free and GitHub Free for organizations. Protected branches are also available in public and private repositories with GitHub Pro, GitHub Team, GitHub Enterprise Cloud, and GitHub Enterprise Server. See [GitHub's plans](#).

Git sanity checklist

All your projects MUST live on git → [BiGHT - Course projects GitHub](#)
This checklist is grade relevant.

1. Protect the main branch. Development happens on feature branches
2. Branches must be given a semantic name



dev_lars_rangein...



21 Branches

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draft: Dev/lars/fa2layout #16 



Merged

lhk merged 14 commits into `main` from `dev/lars/fa2layout`  on Nov 24, 2025

Git sanity checklist

All your projects MUST live on git → [BiGHT - Course projects GitHub](#)

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This is a shared responsibility. You must empower your teammates to collaborate with you.

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6. Data, model weights/checkpoints, .env files, keys don't go in git, write .gitignore on day 1

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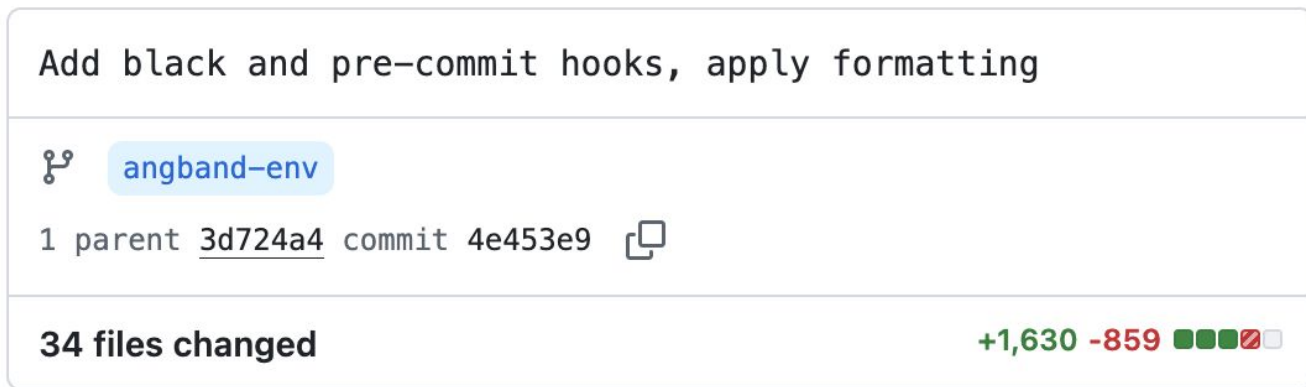
```
[(base) lklein@Larss-MacBook-Pro push2whisper % git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

Code Hygiene

We do expect you to review each other's PRs.

But no one wants to review something like this:



Code Hygiene

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What was the problem in the previous message?

→ 34 changes files, ~1600 lines of code added, ~800 removed!

You can't possibly expect that people read this.

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Ai coding agents will happily change code back and forth, introduce trivial formatting changes, tweak comments, etc

→ Ultimately touching hundreds or thousands of lines of code for meaningless changes

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Your commits **MUST**
be semantic.

Each change is **self
contained**, each
change is **justified**,
each change is
documented.

Code Hygiene

YAML

 Copy

```
repos:
  - repo: https://github.com/astral-sh/ruff-pre-commit
    rev: v0.16.1
    hooks:
      - id: ruff-check
        args: [--fix]
      - id: ruff-format
```

Reproducibility

Did you ever wonder:

- How did this work?
- What numbers went into this plot?
- Why can I not get the same results again?
- What if patient-facing decisions depend on your numbers?

Reproducibility

Did you ever wonder:

[Nature : 1,500 scientists lift the lid on reproducibility](#)

- How did this work?
- What numbers went into this plot?
- Why can I not get the same results again?
- What if patient-facing decisions depend on your numbers?

Survey sheds light on the 'crisis' rocking research.

More than 70% of researchers have tried and failed to reproduce another scientist's experiments, and more than half have failed to reproduce their own experiments. Those are some of the telling figures that emerged from *Nature's* survey of 1,576 researchers who took a brief online questionnaire on reproducibility in research.

Reproducibility

So many reasons reproducibility can fail ...

November 19, 2021

Builds Today, Breaks Tomorrow: The Mystery of the Disappearing NPM Dependency

Bella Wiseman, Lead Engineer, Open Source Program Office

Reproducibility

So many reasons reproducibility can fail ...

```
gatsby
|_gatsby-cli
  |_gatsby-recipes
    |_@graphql-tools/schema
      |_value-or-promise
        |_@changesets/cli
          |_tty-table
            |_smartwrap
```

 GPL

Reproducibility

Many pieces to this puzzle.

But one may be particularly important:

Keep very careful track of all the parameters that go into an experiment!

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Easier said than done. For dependencies there's lockfiles.

Reproducibility

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But one may be particularly important:

Keep very careful track of all the parameters that go into an experiment!

`package-lock.json` is automatically generated for any operations where npm modifies either the `node_modules` tree, or `package.json`. It describes the exact tree that was generated, such that subsequent installs are able to generate identical trees, regardless of intermediate dependency updates.

This file is **intended to be committed** into source repositories, and serves various purposes:

Reproducibility

Many pieces to this puzzle.

But one may be particularly important:

Keep very careful track of all the parameters that go into an experiment!

Easier said than done. For dependencies there's lockfiles.

But how do you track logic, configs, etc?

Reproducibility

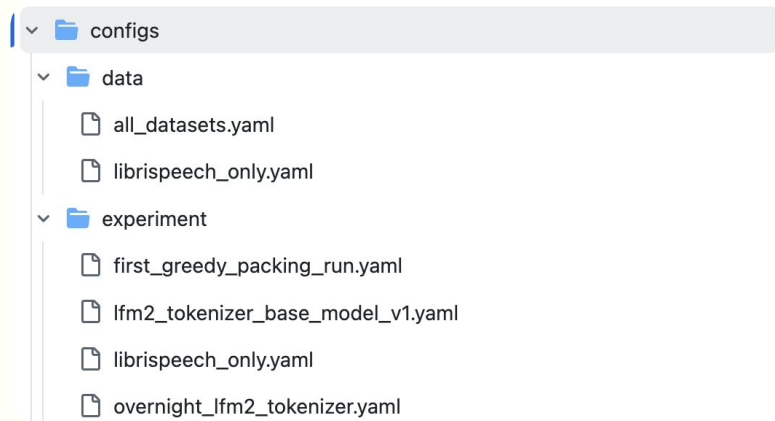
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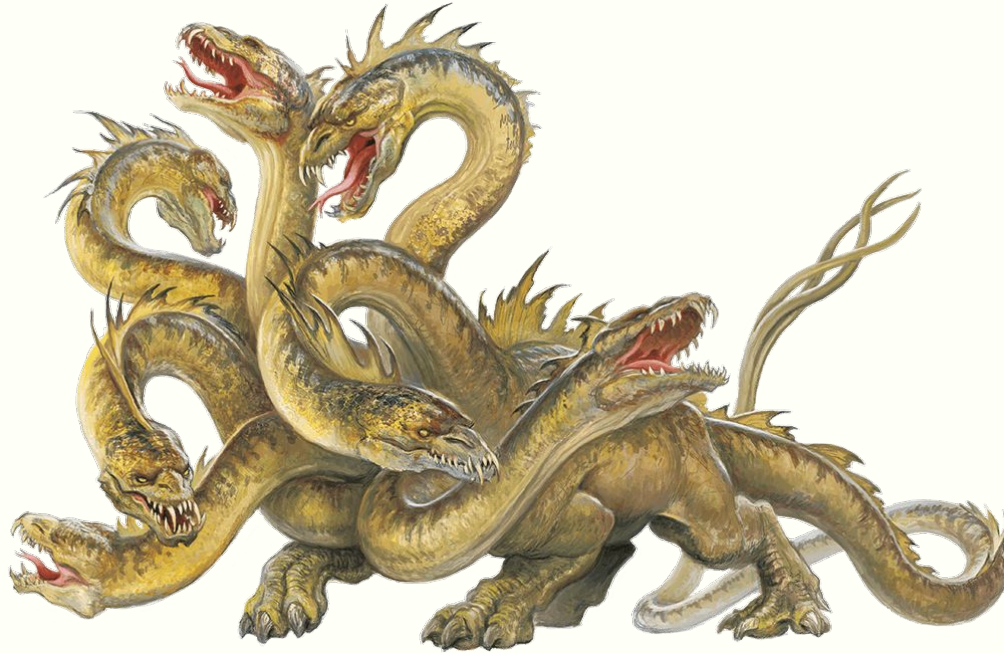
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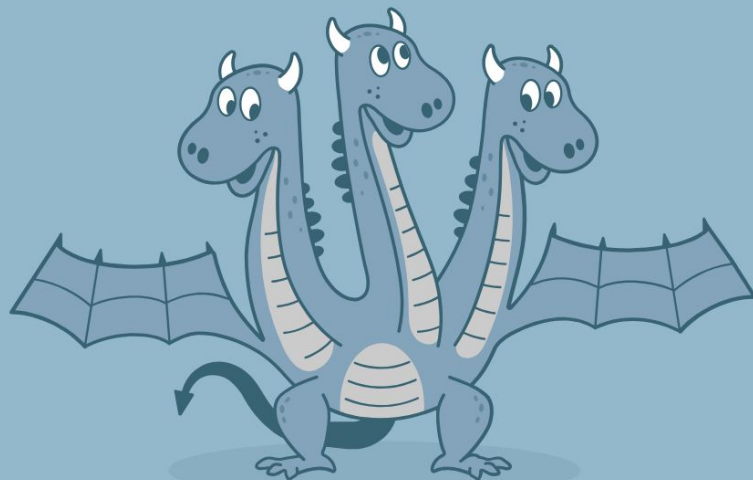
But how do you track logic, configs, etc?



Hydra



Also Hydra (only mildly less scary when first encountered)



Hydra

A framework for elegantly configuring complex applications

[Get Started](#)



Star

10,602

Tracking & Logging

If a tree falls in a forest and no one is around to hear it, does it make a sound?

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If a tree falls in a forest and no one is around to hear it, does it make a sound?

If code runs on the cluster and no one is around to log it, does it waste compute?

Tracking & Logging

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If code runs on the cluster and no one is around to log it, does it waste compute?

Golden Rule:

You **MUST** account for the resources you use.

→ Every number you produce, **MUST** be reproducible. If it isn't, the compute was wasted!

But how?

W&B

- **Unified dashboard:** Central repository for all your model metrics and predictions.
- **Lightweight:** No code changes required to integrate with Hugging Face.
- **Accessible:** Free for individuals and academic teams.
- **Secure:** All projects are private by default.
- **Trusted:** Used by machine learning teams at OpenAI, Toyota, Lyft, and more.

W&B

W&B works like GitHub for machine learning models. Save machine learning experiments to your private, hosted dashboard. Experiment with the confidence that W&B saves all versions of your models, no matter where you run your scripts.

W&B lightweight integrations work with any Python script. Sign up for a free W&B account to start tracking and visualizing your models.

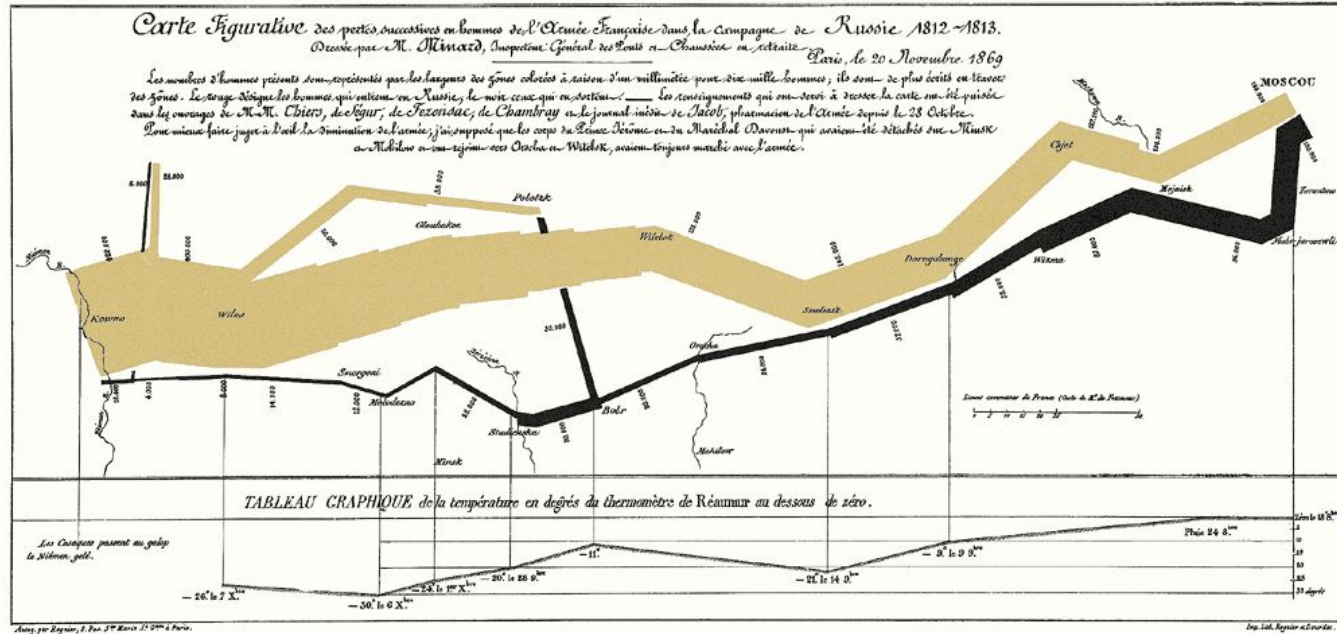
In the Hugging Face Transformers repository, W&B has instrumented the Trainer to automatically log training and evaluation metrics to W&B at each logging step.

W&B



Back to the basics

The art of visual storytelling



```
1 import seaborn as sns
2 from scipy.stats import linregress
3
4 df = sns.load_dataset("penguins").dropna(subset=["bill_depth_mm",
5 "body_mass_g"])
6 fit = linregress(df["bill_depth_mm"], df["body_mass_g"])
7 print(f"n={len(df)}   slope={fit.slope:.1f} g/mm   p=
8 {fit.pvalue:.1e}")
```

→ **n**=342

slope=-191.6 g/mm

p=2.3e-20

```
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```

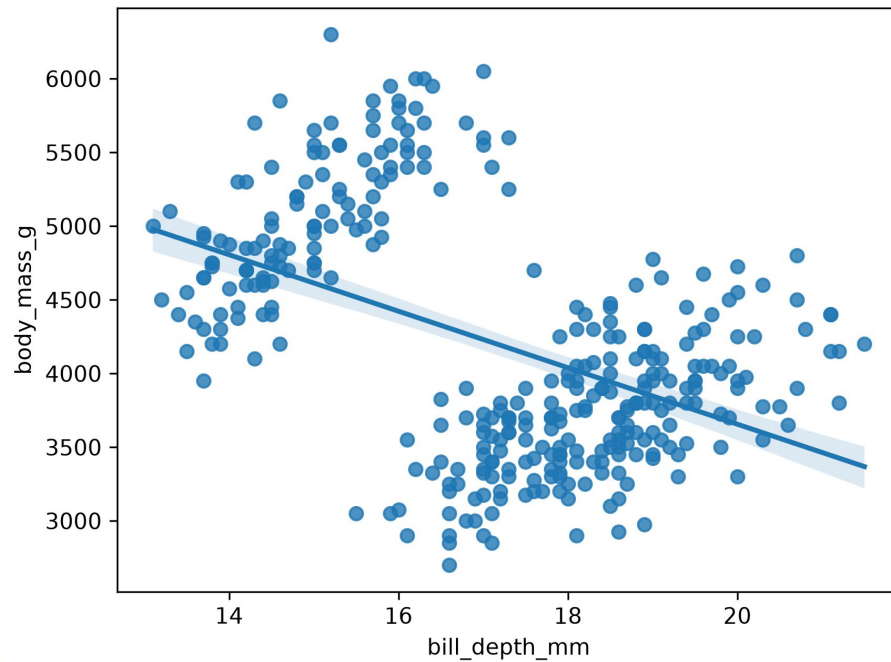
→ n=342

slope=-191.6 g/mm

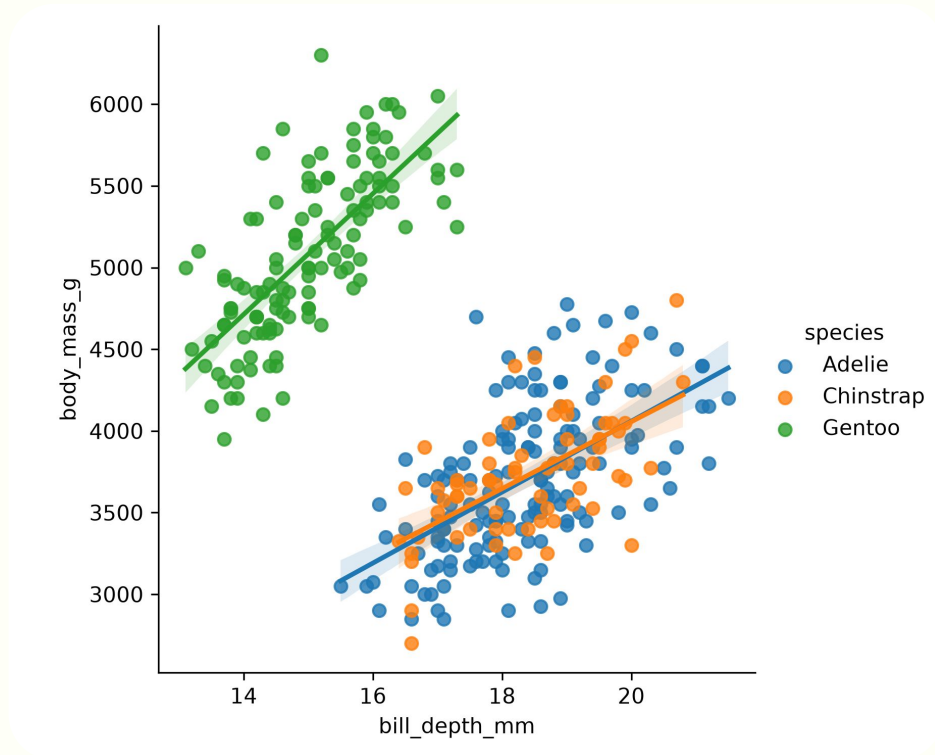
p=2.3e-20

Who here is good with p values?

Scatterplot is your best friend



Scatterplot is your best friend



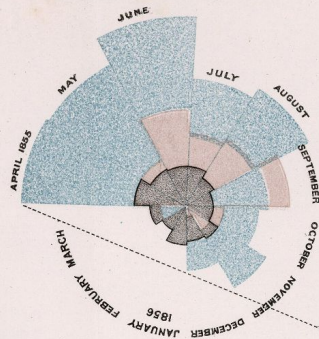
Gentoo is also a penguin (gentoo.org)



And a quick detour to a wonderful website

[Tylervigen - Spurious correlations](#)

2.
APRIL 1855 to MARCH 1856.



The Areas of the blue, red, & black wedges are each measured from the centre as the common vertex.

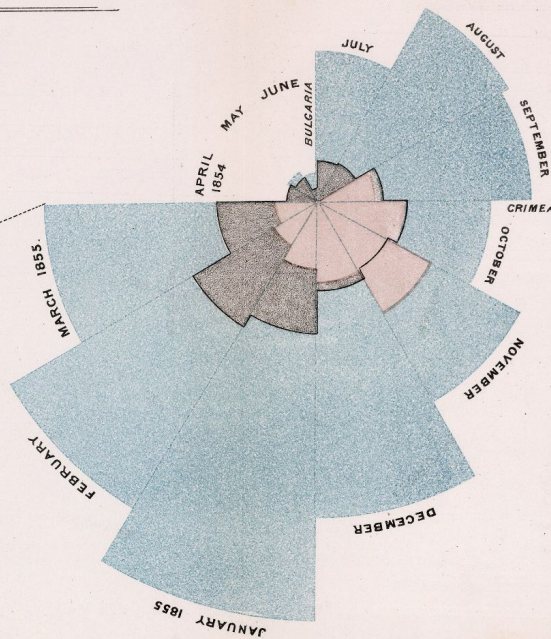
The blue wedges measured from the centre of the circle represent area for area the deaths from Preventible or Mitigable Zymotic diseases, the red wedges measured from the centre the deaths from wounds, & the black wedges measured from the centre the deaths from all other causes.

The black line across the red triangle in Nov^r 1854 marks the boundary of the deaths from all other causes during the month.

In October 1854, & April 1855, the black area coincides with the red; in January & February 1855 the blue coincides with the black.

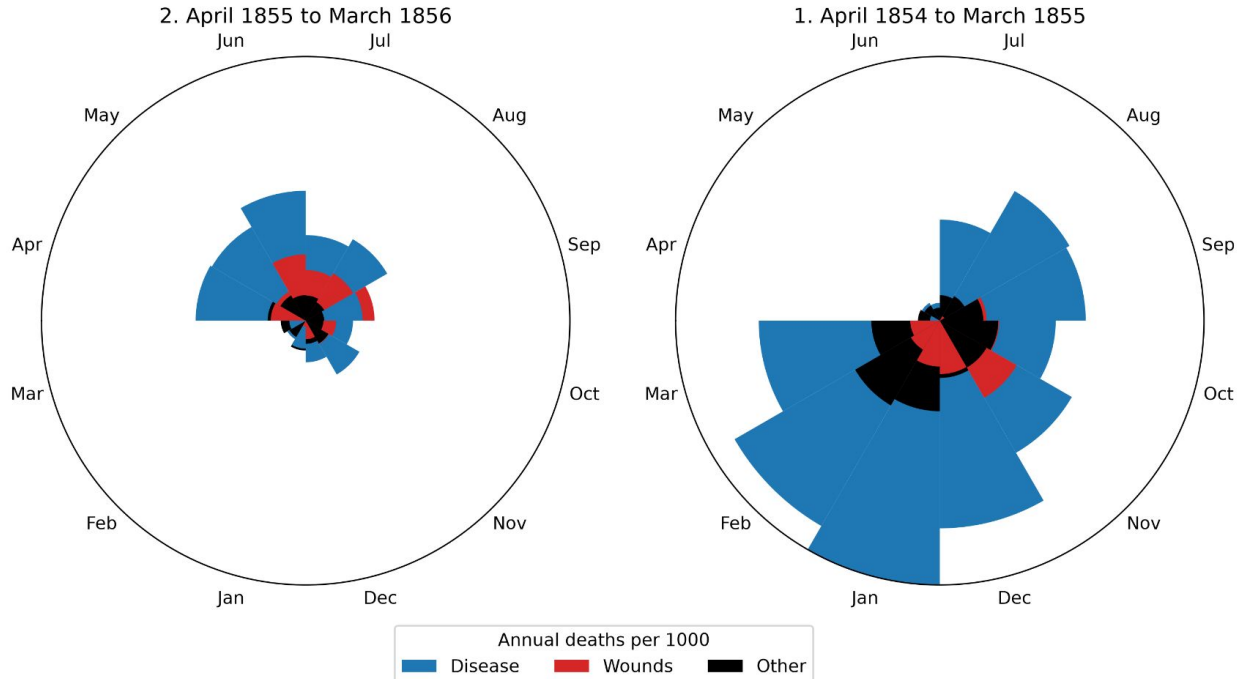
The entire areas may be compared by following the blue, the red & the black lines enclosing them.

1.
APRIL 1854 TO MARCH 1855.

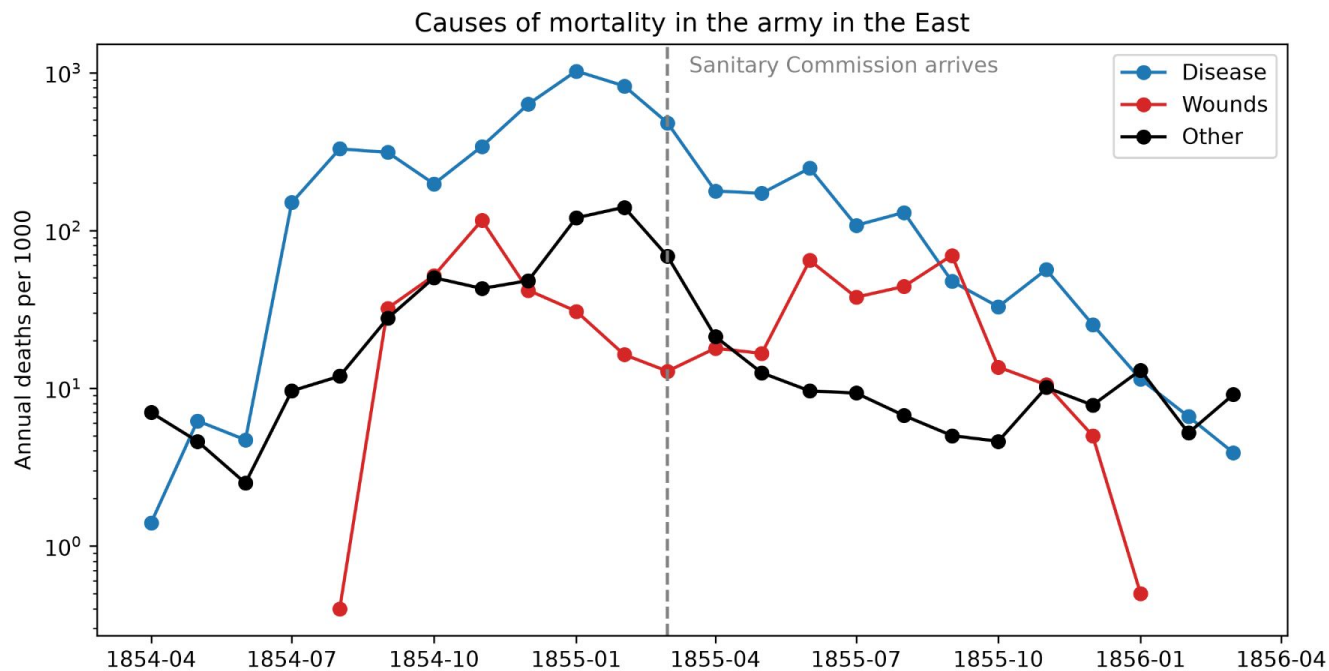


Warrant on J. C. & O. M. & Co. for 1000 lbs.

Diagram of the causes of mortality in the army in the East



Alternative?



Takeaway

Think carefully about the questions you want to ask.

Think carefully about how you can present insightful answers.

Often, simple tooling is enough.

Important to mention here:

- Numpy, pandas
- Matplotlib, seaborn
- Scikit-learn, scipy.stats

Takeaway

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Nature

<https://www.nature.com> › review articles



Array programming with NumPy

by CR Harris · 2020 · Cited by 36644 — NumPy is a community-developed, open-source library, which provides a multidimensional Python array object along with array-aware functions that ...

Before you leave ...

Take a look at the course website.

Once you start working on the projects:

Keep a decision log!

Milestone 1:

Take the time to really think this through!

We expect a careful, in-depth plan. A document that is precise enough to guide your activities throughout the semester. Think about user stories, decision points, risks, fallback scenarios, etc.

And a wonderful tool to help you learn:

Explique!

Thank You

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